



Patient Profile

Patient ID / Name

VPD-2024-001

Age

55

-

+

Biological Sex

Male



Clinical History

Symptom Duration (months)

6



Family history of cancer



Current / former smoker



Analysis Options



Show signaling pathways



Show therapeutic targets



Show screening protocols



PyMutScan —
Virtual Patient
Diagnosis

Gene Panel Analysis · Cancer Staging · Risk
Management · Clinical Recommendations



Symptom Input



Gene Panel



Diagnosis & Staging



Clinical Diagnostic Report



PyMutScan Virtual
Patient Diagnostic
Report

Date/Time: 2026-07-05 13:00

Patient ID: VPD-2024-001 | Age: 55y | Sex: Male

Report Type: AI-Assisted Oncological Risk Assessment

1. PRIMARY DIAGNOSIS

Breast Cancer — Stage IV

5-Year Survival Estimate: 10–25%

Clinical Urgency: Critical

Overall Risk Level: Critical

2. DIFFERENTIAL DIAGNOSES

Rank	Cancer Type	Relative Risk Score
1	Breast	(56)
2	Ovarian	(56)
3	Prostate	(27)
4	Brain	(19)
5	Leukemia	(19)

3. PRESENTING SYMPTOMS (1 recorded)

- Unexplained weight loss (>10%)

Duration: 6 months

Family History: No

Smoking Status: No

4. MUTATIONAL GENE PROFILE (2 genes mutated)

- TP53 (Tumor Suppressor) — Pathway: DNA Damage Response / Apoptosis
- BRCA1 (Tumor Suppressor (DNA Repair)) — Pathway: Homologous Recombination

5. CANCER STAGING ASSESSMENT

Stage IV — Metastatic — distant organ spread confirmed or highly probable.

Criteria	Assessment
Primary Tumor	Present
Lymph Node Spread	Likely (Stage III–IV)
Distant Metastasis	Probable (Stage IV)
Molecular Subtype	Standard genomic profile

6. RISK MANAGEMENT PLAN

 **Emergency Multidisciplinary Board:** Same-day MDT review; initiate treatment protocol immediately

 **Immunotherapy / Precision Therapy:** PD-L1, MSI-H, TMB assessment for checkpoint inhibitor eligibility

 **Clinical Trial Enrollment:** Screen for Phase I/II trials; contact NCI-designated cancer centers

 **Palliative + Curative Dual Approach:** Integrate palliative care from day 1; assess surgical resectability

 **Intensive Biomarker Monitoring:** Weekly ctDNA + monthly imaging for rapid treatment adaptation

 **Holistic Care Team:** Pain management, nutritional, psychological, and spiritual care coordination

 **Advance Care Planning:** Discuss prognosis, goals of care, and advance directives with patient/family

7. RECOMMENDED NEXT STEPS

1. Urgent referral to oncology specialist (Critical priority)

2. Full NGS panel if not yet performed (tumor + germline)
3. Advanced imaging: PET-CT / MRI for staging confirmation
4. Tissue biopsy for histopathological confirmation
5. Multidisciplinary Tumor Board review
6. Patient counseling and informed consent for treatment pathway



Disclaimer: This report is generated by an AI-assisted bioinformatics tool for educational and research purposes only. It does NOT constitute a clinical diagnosis. All findings must be validated by a qualified medical professional.

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